



Mechanism-informed friction-dynamics coupling GRU neural network for real-time cutting force prediction

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ABSTRACT

Monitoring cutting force can provide the most reliable reference information for sensing machining state and supporting adaptive machining. The indirect monitoring approach based on real-time cutting force prediction using Computer Numerical Control (CNC) inherent servo signals has great potential to achieve long-term low-cost and accurate cutting force monitoring since no additional sensors are required. However, due to the complexity of the nonlinear dynamic relationship between the cutting force and servo signals, accurate real-time prediction of cutting force remains an open problem to be investigated in depth. To overcome this challenge, the excitation-response relationship of the machine tool feed system disturbed by cutting force is analyzed first, and then the real-time cutting force prediction problem based on CNC inherent servo signals is defined as a nonlinear dynamic system modeling problem with adaptive time delay. On this basis, a novel mechanism-informed Friction-Dynamics coupling Gated Recurrent Unit (FD-GRU) neural network is proposed. The proposed FD-GRU neural network has two sub GRU neural networks for nonlinear dynamic friction estimation and cutting force inverse estimation respectively, and the mechanism relationship among all related variables is structurally preserved. In addition, a Kalman filter for estimating acceleration is embedded in the part of cutting force inverse estimation to provide more effective input information. Comparative verification was carried out through a set of hole milling experiments containing different combinations of cutting parameters. Compared with the purely data-driven ordinary GRU neural network, the proposed FD-GRU neural network can improve the prediction accuracy and generalization performance by nearly 30% and have better training stability. More importantly, it also can learn the clear mechanism meaning, which provides a strong explanation for its excellent performance.

1. Introduction

Computer numerical control (CNC) machining is widely used in manufacturing processes for various parts in high-end industries such as aerospace, rail transportation, automobiles, and ships. To satisfy the increasing requirements of product performance, monitoring machining states and adaptively controlling the machining process are necessary to ensure the high-quality and efficient machining of parts [1–4]. Cutting force is highly sensitive and can respond quickly to changes in cutting states, making it the most

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directly monitorable physical quantity to reflect cutting states [5–7]. On the one hand, abnormal cutting states such as tool breakage or chatter can be effectively monitored based on the monitored cutting force [8–11]. On the other hand, the monitored cutting force can provide a reliable reference for adaptive machining, so as to maximize the machining efficiency while ensuring the machining quality [12,13].

Monitoring cutting force by force-sensing elements is the most direct and effective technical solution. Currently, table dynamometers, rotary dynamometers, and spindle-integrated sensors are three typical approaches. Table dynamometers can accurately measure cutting force, but the way they are installed and used limits the workpiece size and makes the clamping scheme more complex, which inevitably increases the risk during machining. Rotary dynamometers have no limitations on the workpiece size or clamping scheme but require specialized toolholders. Besides, the measurement coordinate system of rotary dynamometers rotates with the spindle, making it difficult to obtain the feed axis cutting force. On top of that, both types of dynamometers are very expensive. Although spindle-integrated sensors are cheaper, the spindle system needs to be modified to install them, which means it is very difficult to maintain. In addition, force-sensing elements in all three approaches bear the load, so their reliability inevitably decreases under long-term harsh working conditions [14,15]. Hence, although current cutting force monitoring approaches based on force-sensing elements are applicable in laboratory environments, they are difficult to apply to actual machining processes that require higher reliability, practicality, and cost-effectiveness.

From the perspective of motion control of the machine tool feed system, the feed axis cutting force is a type of external disturbance. The motion states of the machine tool feed system are determined by the drive torque of the motor, the friction force as the internal disturbance, the cutting force as the external disturbance, and the structural dynamics of the feed system together. Therefore, it is theoretically possible to reconstruct the cutting force by monitoring two types of servo signals, i.e., drive torque (or drive current) and motion states [15–17]. More importantly, these servo signals are exactly the necessary monitoring signals for motion control of the feed system. In other words, they are inherent signals in CNC systems, which means no additional sensors are needed to collect them. Although these servo signals cannot be directly obtained for some machine tools whose CNC system has bad openness, they still can be monitored through Hall sensors, encoders and grating rulers, etc. On the one hand, these sensors do not invade the workspace, so the normal operation of the machine tool will not be interfered with. On the other hand, they do not directly bear the load, so their reliability is not easily reduced under the long-term harsh working conditions. Therefore, the indirect monitoring method based on real-time cutting force prediction using servo signals has great potential to be applied to actual machining processes that require high reliability, practicality, and cost-effectiveness. According to different modeling ideas, the existing research can be divided into three categories: mechanism-based methods, data-driven methods, and hybrid-driven methods, and they will be reviewed in the following three paragraphs.

Mechanism-based methods predict the interested cutting force component by establishing a mechanism model describing the relationship between servo signals and the cutting force, and the model parameters are identified by some well-designed experiments. For the machine tool feed axis in the form of a ball screw, Yamada et al. [18,19] modeled the feed system as a dual-inertia model and then constructed a disturbance observer to estimate the cutting force. However, the dual-inertia model can only characterize the first-order dynamics, which can not effectively capture the complexity of actual feed systems. Based on the dual-inertia model-based disturbance observer, Yamato et al. [20] constructed a linear filter by experiments to compensate for the prediction results. Aslan et al. [21] established a single-inertia model and nonlinear dynamic friction model by the experiments in non-machining conditions, identified the frequency response function between the cutting force and the responded cutting torque by impact tests, and then constructed an expanded disturbance Kalman filter to predict the cutting force. However, due to the complex characteristics of the servo control system response, nonlinear dynamic friction, and structural dynamics, it is very difficult to accurately predict the cutting force through a pure mechanism model. In addition, the system characteristics identified when the machine tool is idle may be significantly different from the system characteristics when the machine tool is machining. In summary, due to the complexity of the nonlinear dynamic relationship between the cutting force and servo signals, the prediction accuracy of mechanism-based methods is not satisfactory, especially for some complex machining conditions.

Data-driven methods directly establish the end-to-end mapping from servo signals to the cutting force by neural networks based on cutting experiment data. Compared with mechanism-based methods, data-driven methods have a higher upper limit on accuracy. As early as 1999, Kim et al. [22] have attempted to predict the cutting force using a simple static feedforward neural network. Denkena et al. [23] predicted the cutting force using a Long Short-Term Memory (LSTM) neural network. They took servo signals of multiple axes as inputs together and predicted multiple feed axis cutting forces together, and the final prediction performance was not satisfactory. Peng et al. [24] used the root mean square of the spindle three-phase current as the input to predict X-axis and Y-axis cutting forces by a hybrid model consisting of a Convolutional Neural Network (CNN) and an LSTM neural network, and they further integrated a temporal CNN in the model in the subsequent research [25]. However, the spindle current could only partially reflect the information of the feed axis cutting force. Based on the mechanism analysis in [21], Cheng et al. [26] defined the real-time cutting force prediction problem as a nonlinear dynamic system modeling problem with the determined time delay and introduced the Nonlinear Auto-Regression with Exogenous Inputs (NARX) neural network as the prediction model. Ni et al. [27] also adopted this problem definition and further proposed a novel nonparametric-estimation-based unsupervised domain adversarial regression method for cutting force prediction at varying spindle speeds. Although the above-mentioned methods have good potential, purely data-driven methods are considered to rely on a large amount of training data, and have poor interpretability and relatively poor generalization performance [28].

Aiming at the shortcomings of the existing mechanism-based methods and data-driven methods, more and more scholars are trying to adopt the idea of hybrid-driven modeling, so as to retain their respective advantages, while avoiding their respective disadvantages [29–32]. By the way, although there have been many successful applications following the idea of hybrid-driven modeling in the

manufacturing field, the differences among mechanisms of different cases, make hybrid approaches different and usually designed case by case [33–35]. For the real-time cutting force prediction problems, Heydarzadeh et al. [36] proposed a cutting force prediction method for linear motor feed systems using a data-driven residual model to compensate for the incomplete mechanism model. However, a simple rigid-body model is used as the mechanism part in this method, and the residual model is only a static feedforward neural network, which cannot effectively compensate for the complex residual dynamics. Cheng et al. [26] proposed a mechanism-based structured neural network based on the basic architecture of NARX neural networks. The structure of the NARX neural network was modified by taking sub neural networks as approximators for sub models of the mechanism model and reserving the connection relationship among the variables within it. However, it is difficult to determine appropriate time delay orders, and the adopted mode, i.e., single-step training and iterative prediction, is prone to instability when predicting long sequences. Besides, only a static neural network is used in the friction part, which cannot capture the dynamic friction characteristics. In summary, although the methods based on the idea of hybrid-driven modeling have great potential to further improve the prediction accuracy, generalization performance, and interpretability of data-driven methods, existing research still has much room for improvement.

To achieve accurate real-time cutting force prediction based on CNC inherent servo signals, a novel mechanism-informed Friction-Dynamics coupling Gated Recurrent Unit (FD-GRU) neural network is proposed following the idea of hybrid-driven modeling and verified in this paper. The main contributions are summarized as follows.

- 1) By analyzing the excitation-response relationship of the machine tool feed system disturbed by cutting force, the real-time cutting force prediction problem based on CNC inherent servo signals is defined as a nonlinear dynamic system modeling problem with adaptive time delay. On this basis, the purely data-driven ordinary Gated Recurrent Unit (GRU) neural network is introduced as the real-time cutting force prediction model. It is not only the basic architecture of the proposed FD-GRU neural network, but also taken as the baseline model for comparative verification.
- 2) Following the idea of hybrid-driven modeling, a mechanism-informed friction-dynamics coupling GRU neural network for real-time cutting force prediction is further proposed. Two sub GRU neural networks are introduced to represent the parts of nonlinear dynamic friction estimation and cutting force inverse estimation respectively, and the mechanism relationship among all related variables is structurally preserved. In addition, a Kalman filter for estimating acceleration is embedded in the part of cutting force inverse estimation to provide more effective input information.
- 3) A dataset was established through a set of hole milling experiments containing combinations of different spindle speeds, axial depths, radial depths, and feedrates for comparative verification. Experimental results show that compared with the purely data-driven ordinary GRU neural network, the proposed FD-GRU neural network can improve the prediction accuracy and generalization performance by nearly 30 %, and have better training stability. In addition, it also can learn the clear mechanism meaning, which provides a strong explanation for its excellent performance.

The rest of this paper is organized as follows. In Section 2, the principle and problem definition of real-time cutting force prediction are presented, which is the basis for designing the following prediction models. In Section 3, the ordinary GRU neural network is introduced as the real-time cutting force prediction model, which is also the baseline model in this paper. In Section 4, following the idea of hybrid-driven modeling, a novel FD-GRU neural network is designed for real-time cutting force prediction. In Section 5, the comparison verification using real experimental data is shown. Finally, the conclusion is drawn in Section 6.

2. Principle and problem definition of real-time cutting force prediction

Ball-screw drives are widely used in machine tools as the main element of the feeding system [37]. Without loss of generality, the machine tool feed system in the form of a ball screw is analyzed as an example in this section. As shown in Fig. 1(a), the drive torque outputted by the servo motor acts on the ball screw, which in turn generates acceleration and rotation, and drives the worktable

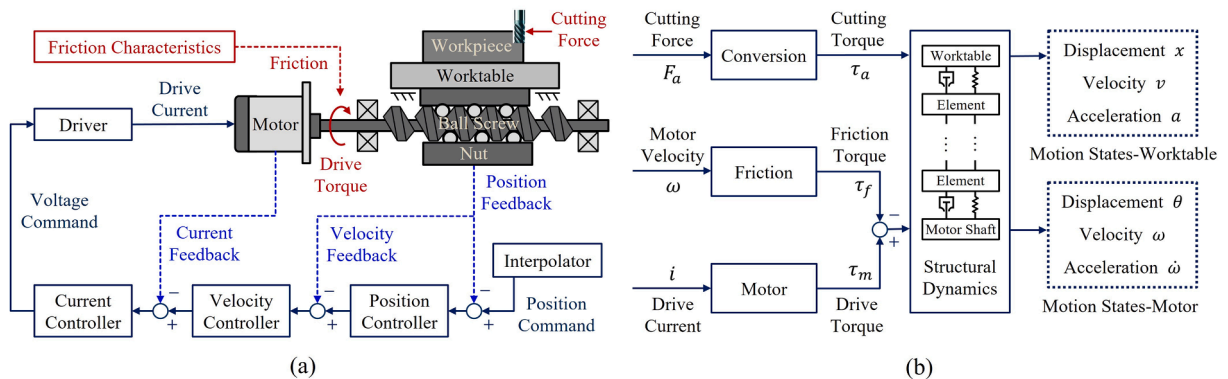


Fig. 1. (a) Diagram of the closed-loop control of the machine tool feed system disturbed by the cutting force. (b) The excitation-response relationship of the machine tool feed system disturbed by the cutting force.

through the nut. At the same time, the cutting force acts on the worktable as an external disturbance and affects the motion states of the feed system with the drive torque together. In addition, the impact of the friction characteristics inside the feed system on motion states cannot be ignored. During the control process, servo signals such as the displacement and velocity of the worktable, the drive current (drive torque), and the motor velocity, are all fed back to the closed-loop control system through sensors. The control system then adjusts the output command according to the feedback variables to track the input command. Hence, when the cutting force acts on the worktable, its information will be reflected in the CNC inherent servo signals such as the drive current of the motor and the motion states of the feed system, which makes it possible to reconstruct the cutting force by them.

The interaction mechanism between CNC inherent servo signals and cutting forces is relatively complex, including many links such as machinery, motors, and control, and showing complex nonlinear dynamic characteristics. To establish the real-time cutting force prediction model based on CNC inherent servo signals, the relationship between the cutting force and servo signals is analyzed first from the perspective of the excitation-response relationship. As shown in Fig. 1(b), the feed axis cutting force (F_a) is converted into the cutting torque τ_a acting on the worktable end of the feed system through transmission. At the motor end of the feed system, the drive current i drives the motor to generate the drive torque τ_m . It should be noted that for permanent magnet synchronous motors, which are commonly used in machine tool feed systems, the drive current is exactly the torque current. In addition, the friction characteristics inside the feed system are usually equivalent to the friction torque τ_f that acts on the motor end and exists in the form of motor rotation resistance torque. Therefore, the machine tool feed system is excited by the drive torque and friction torque at the motor end and the cutting torque at the worktable end together, and the motion states (s) are the final response of the structural dynamics. The motion states of the feed system include the displacement θ , velocity ω , and acceleration $\dot{\omega}$, etc., at the motor end, as well as the displacement x , velocity v , and acceleration a etc., at the worktable end.

Specifically, drive torque τ_m is the product of the drive current i and the motor torque constant K :

$$\tau_m = Ki \quad (1)$$

Cutting torque τ_a can be calculated by the cutting force in the direction of the feed axis (F_a). The conversion relationship between the cutting force and the cutting torque is expressed as:

$$\tau_a = F_a \frac{h_p}{2\pi\eta} \quad (2)$$

where h_p and η are the pitch length and the transmission efficiency of the ball screw, respectively.

The machine tool feed system has various static and dynamic friction characteristics, such as Coulomb friction, viscous friction, static friction, Stribeck effect, pre-sliding displacement, and frictional hysteresis, etc. In order to accurately model these friction characteristics, LuGre model, a nonlinear dynamic friction model, can be used [38]. The LuGre model assumes that the two microscopic contact surfaces are a large number of elastic bristles with random behavior, and is modeled based on the average deformation of the bristles (z). The friction torque τ_f of the LuGre model is calculated by:

$$\tau_f = \sigma_0 z + \sigma_1 \left(\frac{dz}{dt} \right) + \sigma_2 \omega \quad (3)$$

$$\frac{dz}{dt} = \omega - \frac{\sigma_0}{g(\omega)} z |\omega| \quad (4)$$

$$g(\omega) = \tau_c + (\tau_s - \tau_c) e^{-\left(\frac{\omega}{\omega_s} \right)^2} \quad (5)$$

where ω denotes the velocity, i.e., the motor velocity of the feed system. σ_0 and σ_1 denotes the stiffness and microcosmic damping coefficients between two contact surfaces, respectively. σ_2 denotes the macroscopical viscous friction coefficient. $g(\omega)$ is the Stribeck effect function, where τ_c denotes the Coulomb friction torque, τ_s denotes the static friction torque, and ω_s denotes the Stribeck velocity. It should be noted that the average deformation of the bristles (z) is a hidden state that cannot be directly observed, and the observable variable is only the motor velocity ω . Therefore, z is usually calculated iteratively from the zero state. Due to the discreteness of the actual monitoring process, the LuGre model shown in Eqs. (3–5) can be expressed as a discrete-time nonlinear dynamic function with the motor velocity ω as the input in a general form (F_f):

$$\tau_f(t) = F_f \left(\begin{array}{c} \tau_f(t-1), \dots, \tau_f(t-q), \\ \omega(t), \omega(t-1), \dots, \omega(t-p) \end{array} \right) \quad (6)$$

where t denotes the current time. p and q denote the time delay orders of the velocity and friction torque, respectively.

In consideration of the structural dynamics of the feed system, the established forward excitation-response model can be expressed as a discrete-time nonlinear dynamic function in a general form (F_{er}):

$$s(t) = F_{er} \left(\begin{array}{c} s(t-1), \dots, s(t-m), \\ (\tau_m(t) - \tau_f(t)), (\tau_m(t-1) - \tau_f(t-1)), \dots, (\tau_m(t-n) - \tau_f(t-n)) \\ F_a(t), F_a(t-1), \dots, F_a(t-l) \end{array} \right) \quad (7)$$

where m , n and l denote the time delay orders of the motion states s , the resultant torque of the drive torque τ_m and the friction torque τ_f , and the feed axis cutting force (F_a), respectively. To predict the cutting force, Eq. (7) can be rewritten as a discrete-time dynamic function in a general form (F_{re}):

$$F_a(t) = F_{re} \left(\begin{array}{c} F_a(t-1), \dots, F_a(t-l), \\ (\tau_m(t) - \tau_f(t)), (\tau_m(t-1) - \tau_f(t-1)), \dots, (\tau_m(t-n) - \tau_f(t-n)) \\ s(t), s(t-1), \dots, s(t-m) \end{array} \right) \quad (8)$$

This equation represents the dynamical estimation of the cutting force using the drive torque, motion states, and friction torque, which is called cutting force inverse estimation in this paper.

By combining Eq. (1) for calculating the drive torque and Eq. (6) for calculating the friction torque, Eq. (8) can be further expressed as:

$$F_a(t) = F_{res} \left(\begin{array}{c} F_a(t-1), \dots, F_a(t-l), \\ \left(Ki(t) - F_f \left(\begin{array}{c} \tau_f(t-1), \dots, \tau_f(t-q), \\ \omega(t), \omega(t-1), \dots, \omega(t-p) \end{array} \right) \right), \\ \left(Ki(t-1) - F_f \left(\begin{array}{c} \tau_f(t-2), \dots, \tau_f(t-1-q), \\ \omega(t-1), \omega(t-2), \dots, \omega(t-1-p) \end{array} \right) \right), \dots, \\ \left(Ki(t-n) - F_f \left(\begin{array}{c} \tau_f(t-n-1), \dots, \tau_f(t-n-q), \\ \omega(t-n), \omega(t-n-1), \dots, \omega(t-n-p) \end{array} \right) \right) \\ s(t), s(t-1), \dots, s(t-m) \end{array} \right) \quad (9)$$

Since the friction torque τ_f can be calculated solely by the motor velocity ω , friction model F_f can be integrated into F_{res} . Therefore, the discrete-time nonlinear dynamic function representing the end-to-end mapping relationship from CNC inherent signals to the cutting force (F_{ete}) can be expressed as:

$$F_a(t) = F_{ete} \left(\begin{array}{c} F_a(t-1), \dots, F_a(t-l), \\ i(t), i(t-1), \dots, i(t-n), \\ \omega(t), \omega(t-1), \dots, \omega(t-o), \\ s(t), s(t-1), \dots, s(t-m) \end{array} \right) \quad (10)$$

where l , n , o and m denote the time delay orders of the cutting force, drive current, motor velocity, and motion states, respectively. It should be noted that since ω has been taken as the input separately, it should no longer be included in s .

According to Eq. (10), to establish the real-time cutting force prediction model is to determine F_{ete} and the corresponding time delay orders. Since F_{ete} is a nonlinear function, if time delay orders are determined, an ordinary neural network can be used to learn F_{ete} directly from the observation data. This form of neural network is called NARX neural network. However, it is difficult to choose appropriate time delay orders. Moreover, the adopted mode of NARX neural networks, i.e., single-step training and iterative prediction, is prone to instability when predicting long sequences. Therefore, the hidden state vector $\mathbf{h}$ of the system is further introduced to transform Eq. (10) into a nonlinear dynamic system modeling problem with adaptive time delay and expressed as:

$$F_a(t) = F_{ete}^{ad}(i(t), \omega(t), s(t), \mathbf{h}(t-1)) \quad (11)$$

where F_{ete}^{ad} denotes the nonlinear dynamic function with adaptive time delay. $\mathbf{h}(t-1)$ denotes the hidden state vector at the previous time $t-1$, which contains the history information of the input and output and is automatically updated with the iterative calculation of F_{ete}^{ad} . Hence, an effective real-time cutting force prediction model can be established by designing an appropriate neural network to represent F_{ete}^{ad} .

3. GRU neural network-based real-time cutting force prediction

According to the mechanism analysis in Section 2, the real-time cutting force prediction problem is defined as a nonlinear dynamic system modeling problem with adaptive time delay. Traditional Recurrent Neural Networks (RNNs) are a type of neural network that can model the nonlinear dynamic system in a discrete-time form without knowing the time delay [39], which has the potential to be taken as the cutting force prediction model. Nevertheless, when the input is a long sequence, traditional RNN is prone to gradient vanishing or explosion due to the excessive retention of data, making the model difficult to train and apply [40,41]. For the cutting force prediction task, because the sampling frequency of CNC inherent servo signals is hundreds or thousands of Hz, so the input sequence is relatively long in the training process. To overcome this challenge, two variants of the traditional RNN, i.e., Long Short-Term Memory (LSTM) neural network and Gated Recurrent Unit (GRU) neural network were proposed by scholars. Through the specially designed gate structures, they both have adaptive long short-term memory ability, making them suitable for long sequence prediction problems [41,42]. Compared with the LSTM neural network, the GRU neural network has fewer parameters and higher computational efficiency with almost the same performance [42]. Hence, the GRU neural network is introduced as the real-time

cutting force prediction model in this paper.

GRU neural network-based real-time cutting force prediction is shown in Fig. 2. The main part of the network contains a standard GRU and a fully-connected layer. At the current time t , the GRU neural network receives the hidden state at time $t-1$ ($\mathbf{h}(t-1)$) and the monitored servo signals at time t ($\mathbf{x}(t)$), then predicts the cutting force at time t ($F_a(t)$), and outputs the updated hidden state at time t ($\mathbf{h}(t)$). Specifically, GRU first calculates the output of the reset gate at time t ($\mathbf{R}(t)$) by $\mathbf{h}(t-1)$ and ($\mathbf{x}(t)$):

$$\mathbf{R}(t) = \sigma(\mathbf{W}_{rh}\mathbf{h}(t-1) + \mathbf{W}_{rx}\mathbf{x}(t) + \mathbf{b}_r) \quad (12)$$

where $\mathbf{W}_{rh}$ and $\mathbf{W}_{rx}$ denote the corresponding weight matrices, and $\mathbf{b}_r$ denotes the corresponding bias vector. In addition, the subsequent $\mathbf{W}$ and $\mathbf{b}$ have similar meanings and will not be described again. σ denotes the sigmoid function. Then $\mathbf{R}(t)$ is used to obtain the candidate hidden state at time t ($\tilde{\mathbf{h}}(t)$):

$$\tilde{\mathbf{h}}(t) = \tanh(\mathbf{W}_{hh}(\mathbf{R}(t) \odot \mathbf{h}(t-1)) + \mathbf{W}_{ih}\mathbf{x}(t) + \mathbf{b}_h) \quad (13)$$

where $\odot$ is Hadamard product, i.e., element-wise product.

While calculating $\tilde{\mathbf{h}}(t)$, the output of the update gate at time t ($\mathbf{Z}(t)$) is calculated by $\mathbf{h}(t-1)$ and $\mathbf{x}(t)$:

$$\mathbf{Z}(t) = \sigma(\mathbf{W}_{zh}\mathbf{h}(t-1) + \mathbf{W}_{iz}\mathbf{x}(t) + \mathbf{b}_z) \quad (14)$$

$\mathbf{Z}(t)$ represents the selective forgetting of the original hidden state of the system. The closer the element in $\mathbf{Z}(t)$ is to 0, the more forgotten the corresponding element in the hidden state. However, $1 - \mathbf{Z}(t)$ means adding memory to the candidate hidden state, which is exactly the opposite of $\mathbf{Z}(t)$ and makes it possible to use one gate to simultaneously forget and update. After obtaining $\tilde{\mathbf{h}}(t)$ and $\mathbf{Z}(t)$, $\mathbf{h}(t)$ is calculated by:

$$\mathbf{h}(t) = \tilde{\mathbf{h}}(t) \odot (1 - \mathbf{Z}(t)) + \mathbf{h}(t-1) \odot \mathbf{Z}(t) \quad (15)$$

Finally, $F_a(t)$ is obtained from $\mathbf{h}(t)$ through a fully-connected layer. A linear fully-connected layer is used in this paper, so the transformation from $\mathbf{h}(t)$ to $F_a(t)$ is expressed as:

$$F_a(t) = \mathbf{W}_f\mathbf{h}(t) + \mathbf{b}_f \quad (16)$$

Denoting the GRU neural network as GRU_{NN} , the GRU neural network-based real-time cutting force prediction can be expressed as:

$$F_a(t) = \text{GRU}_{\text{NN}}(\mathbf{x}(t), \mathbf{h}(t-1)) \quad (17)$$

where the input vector $\mathbf{x}(t) = [i(t), \omega(t), s^T(t)]^T$. The hidden state at time $t-1$ ($\mathbf{h}(t-1)$) contains the adaptively learned information at past times, and is determined by iterative calculation from the initial state. Therefore, it is no longer necessary to artificially specify the time delay orders of the input and output.

The sequence-to-sequence approach is used for supervised training, to improve the stability of the GRU neural network when predicting long cutting force sequences. In addition, for each training trajectory, the initial hidden state $\mathbf{h}(0)$ should be set to a zero vector to ensure that all trajectories are predicted from the same initial state. Accordingly, the trajectory data of cutting force in the dataset should also start from zero. When applying the trained model, it should also start predicting from the zero initial state continuously. In this paper, Mean Square Error (MSE) loss is chosen as the loss function of the GRU neural network and expressed as:

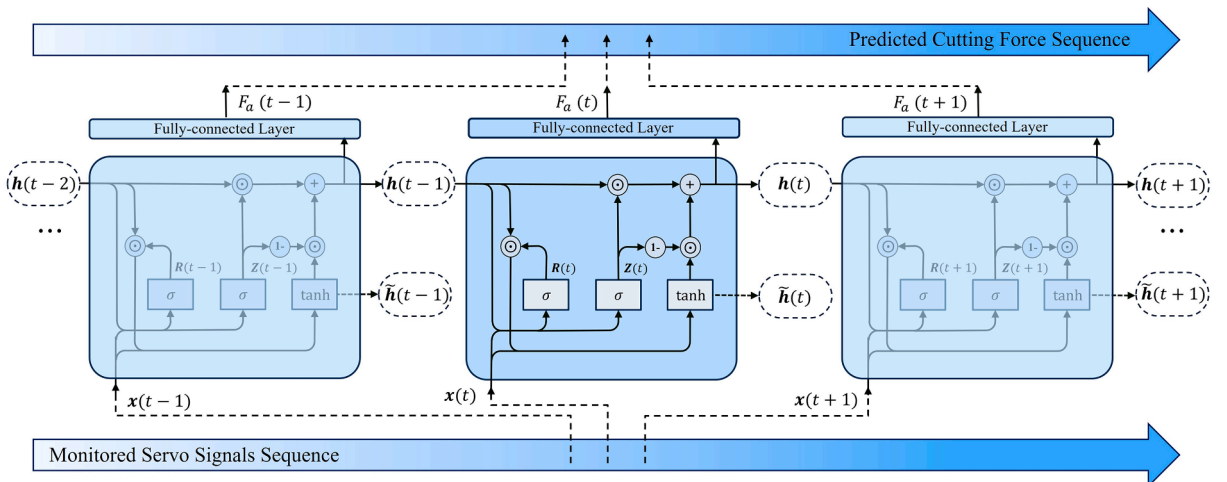


Fig. 2. GRU neural network-based real-time cutting force prediction.

$$\text{MSELoss} = \frac{1}{MN} \sum_{j=1}^M \sum_{t=1}^{N_j} \left(y_j(t) - \text{GRU}_{\text{NN}}(\mathbf{x}_j(t), \mathbf{h}_j(t-1)) \right)^2 \quad (18)$$

where M denotes the number of sequences, j denotes the No of the j -th sequence, and N_j denotes the number of sample times, i.e., the length of the j -th sequence. $y_j(t)$ denotes the label value at time t in the j -th sequence. $\mathbf{x}_j(t)$ denotes the input of the GRU neural network at time t in the j -th sequence. $\mathbf{h}_j(t-1)$ denotes the hidden state of the GRU neural network at time $t-1$ in the j -th sequence.

It is important to note that since the real-time cutting force prediction problem has never been defined before as in Section 2, the GRU neural network or even the LSTM neural network has never been used in the same way as shown in this section. Although the LSTM neural network and its variants have been used for real-time cutting force prediction [23–25], they were just taken as a sequence-to-sequence prediction model without considering the characteristics of dynamic systems. Therefore, they are difficult to capture real dynamic characteristics and cannot predict very well. GRU neural network-based real-time cutting force prediction is not only proposed as a novel method in this paper, but also as the baseline for verifying the great performance of the following proposed FD-GRU neural network.

4. Mechanism-informed FD-GRU neural network

As stated in Section 1, data-driven methods are considered to rely on a large amount of training data, have poor interpretability, and have relatively poor generalization performance. Although the GRU neural network introduced in Section 3 is suitable for the real-time cutting force prediction problem, it inevitably has the inherent disadvantages of data-driven models. In addition, although it is very difficult to construct an accurate pure mechanism-based real-time cutting force prediction model, the mechanism relationship among different variables and the basic characteristics of different sub models are still reliable. Therefore, we design a novel mechanism-informed FD-GRU neural network following the idea of hybrid-driven modeling. As stated in Section 2, the two most complex parts of real-time cutting force prediction are nonlinear dynamic friction estimation and cutting force inverse estimation. In consideration of the nonlinear dynamic characteristics of the two estimation processes, two sub GRU neural networks are introduced to represent them separately. In addition, the mechanism relationship among all related variables is structurally preserved. By this way, the proposed FD-GRU neural network has similar internal structures to the mechanism model, and they can be viewed as two formally equivalent models. On the one hand, since the main body of the proposed FD-GRU neural network is still GRU neural networks, it can inherit their powerful modeling ability. On the other hand, compared with the ordinary GRU neural network with the same model complexity, the introduced mechanism constraints on the network structure reasonably reduce the solution space of the model. Therefore, under the same training settings and model complexity, the proposed FD-GRU neural network has more potential to achieve better prediction accuracy and generalization performance. Besides, the training stability may also have the potential to be improved.

The proposed FD-GRU neural network is shown in Fig. 3. The sub GRU neural network for estimating the nonlinear dynamic friction is called Friction-GRU and is denoted as GRU_f . The sub GRU neural for estimating the cutting force inversely is called Dynamics-GRU and denotes as GRU_d . The input–output relationship of GRU_f can be expressed as:

$$\tilde{\tau}_f(t) = \text{GRU}_f(\omega(t), \mathbf{h}_f(t-1)) \quad (19)$$

where $\tilde{\tau}_f$ denotes the estimated friction torque, and $\mathbf{h}_f$ denotes the hidden state of GRU_f . It should be noted that because all inputs in the FD-GRU neural network are normalized and there is no direct supervision of the output of Friction-GRU, the estimated friction torque is just a relative value without units. Nevertheless, if the Friction-GRU can effectively capture the characteristics of nonlinear dynamic friction, it is still possible to significantly reduce the solution space of the overall model.

Subtract $\tilde{\tau}_f$ from the drive torque, and use it with the motion states together as the input of Dynamics-GRU (GRU_d) to predict the cutting force ($F_d(t)$):

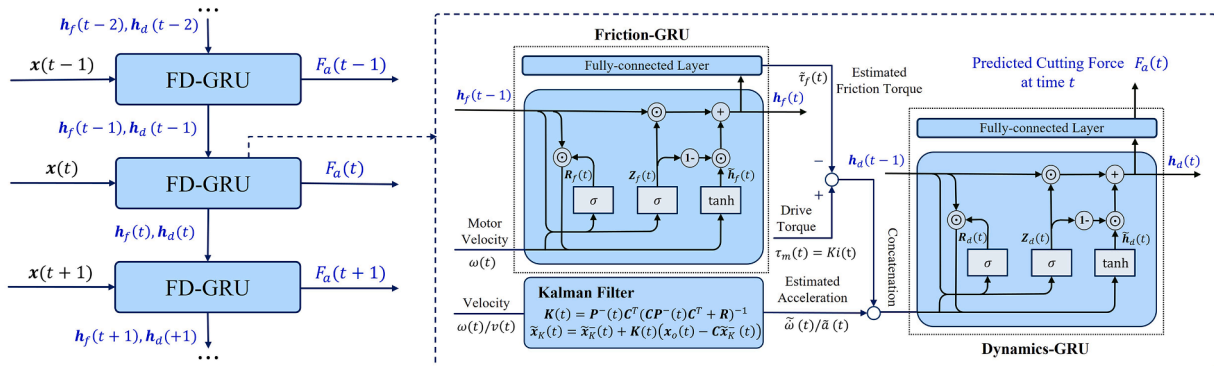


Fig. 3. The architecture of the proposed FD-GRU neural network.

$$F_a(t) = \text{GRU}_d(\mathbf{x}_d(t), \mathbf{h}_d(t-1)) \quad (20)$$

where $\mathbf{h}_d$ denotes the hidden state of GRU_d . $\mathbf{x}_d$ denotes the input of GRU_d , and $\mathbf{x}_d$ at time t ($\mathbf{x}_d(t)$) can be expressed as:

$$\mathbf{x}_d(t) = \left[\left(Ki(t) - \tilde{\tau}_f(t) \right), \mathbf{s}^T(t) \right]^T \quad (21)$$

where $\mathbf{s}$ denotes the selected motion states of the machine tool feed system. Acceleration is more sensitive to the force response than velocity and displacement, which can provide more effective information for predicting the cutting force. However, the motor acceleration or worktable acceleration is not directly monitored in ordinary CNC systems. Nevertheless, the motor velocity and worktable velocity are both CNC inherent servo signals. Therefore, a Kalman filter for estimating acceleration via velocity is embedded in the part of cutting force inverse estimation to provide more effective input information. Without loss of generality, the estimation method of motor acceleration is introduced as an example in this paper.

First, the discrete-time state-space model of the motion states including motor velocity ω and motor acceleration $\dot{\omega}$ is defined as:

$$\mathbf{x}_K(t+1) = \mathbf{A}\mathbf{x}_K(t) + \mathbf{w}(t) \quad (22)$$

$$\mathbf{x}_o(t) = \mathbf{C}\mathbf{x}_K(t) + \mathbf{v}(t) \quad (23)$$

where $\mathbf{x}_K(t)$ denotes the system state vector at time t and is defined as $\mathbf{x}_K(t) = [\omega(t) \quad \dot{\omega}(t)]^T$. $\mathbf{w}$ and $\mathbf{v}$ denote the process noise vector and measurement noise vector, respectively. $\mathbf{A}$ denotes the system state matrix and is defined as:

$$\mathbf{A} = \begin{bmatrix} 1 & dt \\ 0 & 1 \end{bmatrix}$$

where dt denotes the system sampling period. $\mathbf{C}$ denotes the system observation matrix and is defined as a 2×2 unit matrix. $\mathbf{x}_o(t)$ denotes the observation state vector at time t and is defined as:

$$\mathbf{x}_o(t) = \begin{bmatrix} \omega_o(t) & \dot{\omega}_o(t) \end{bmatrix}^T = \begin{bmatrix} \omega_o(t) & (\omega_o(t) - \omega_o(t-1))/dt \end{bmatrix}^T$$

where $\omega_o(t)$ is the measured value of the motor side speed at time t , $\dot{\omega}_o(t)$ is the acceleration directly obtained by the first-order differencing of velocity. On this basis, a standard Kalman filter can be constructed to estimate the motor acceleration [43]:

$$\tilde{\mathbf{x}}_K^-(t) = \mathbf{A}\tilde{\mathbf{x}}_K(t-1) \quad (24)$$

$$\mathbf{P}^-(t) = \mathbf{A}\mathbf{P}(t-1)\mathbf{A}^T + \mathbf{Q} \quad (25)$$

$$\mathbf{K}(t) = \mathbf{P}^-(t)\mathbf{C}^T(\mathbf{C}\mathbf{P}^-(t)\mathbf{C}^T + \mathbf{R})^{-1} \quad (26)$$

$$\tilde{\mathbf{x}}_K(t) = \tilde{\mathbf{x}}_K^-(t) + \mathbf{K}(t)\left(\mathbf{x}_o(t) - \mathbf{C}\tilde{\mathbf{x}}_K^-(t)\right) \quad (27)$$

$$\mathbf{P}(t) = (\mathbf{I} - \mathbf{K}(t)\mathbf{C})\mathbf{P}^-(t) \quad (28)$$

where $\tilde{\mathbf{x}}_K^-(t)$ and $\tilde{\mathbf{x}}_K(t)$ denote the predicted state vector and final estimated state vector at time t , respectively. $\mathbf{Q}$ and $\mathbf{R}$ denote the covariance matrices of process noise and measurement noise, respectively. Eqs. (24–25) are the prediction process. The predicted state vector and new state covariance matrix at time t ($\tilde{\mathbf{x}}_K^-(t)$ and $\mathbf{P}^-(t)$) are calculated from the final estimated state vector and state covariance matrix at time $t-1$ ($\tilde{\mathbf{x}}_K(t-1)$ and $\mathbf{P}(t-1)$). Eqs. (26–28) are the correction process. The final estimated state vector and updated state covariance matrix at time t ($\tilde{\mathbf{x}}_K(t)$ and $\mathbf{P}(t)$) are calculated from the observation state vector and computed Kalman gain matrix at time t ($\mathbf{x}_o(t)$ and $\mathbf{K}(t)$). After obtaining $\tilde{\mathbf{x}}_K(t)$, the final estimated motor acceleration at time t ($\tilde{\dot{\omega}}(t)$) can be determined by:

$$\tilde{\dot{\omega}}(t) = \mathbf{C}_a\tilde{\mathbf{x}}_K(t) = [0 \quad 1]\tilde{\mathbf{x}}_K(t) \quad (29)$$

where $\mathbf{C}_a$ denotes the observation matrix of the estimated motor acceleration.

Based on the above content, the proposed FD-GRU neural network can be expressed as:

$$F_a(t) = \text{GRU}_d\left(\left[(Ki(t) - \text{GRU}_f(\omega(t), \mathbf{h}_f(t-1))), \text{KF}(t)\right]^T, \mathbf{h}_d(t-1)\right) \quad (30)$$

where $\text{KF}(t)$ denotes the motor acceleration at time t estimated by Eqs. (24–29). It can be seen that the proposed FD-GRU neural network can be regarded as a specially structured GRU neural network. On the one hand, it inherits the powerful modeling capability of the GRU neural network for time sequences, ensuring that the model has a great upper limit of accuracy. On the other hand, structural constraints are introduced using the mechanism of real-time cutting force prediction as a priori, which is equivalent to

introducing reasonable inductive bias into the model. As a result, the sub parts of the overall model will be guided to learn different mechanism meanings, reducing the solution space of the overall model. In the following Section 4, experimental results will also demonstrate this point.

Similar to the GRU neural network-based real-time cutting force prediction, the sequence-to-sequence approach is used for training the FD-GRU neural network, and two initial hidden states ($\mathbf{h}_f(0)$ and $\mathbf{h}_d(0)$) should also be set to zero vectors. Mean Square Error (MSE) loss is chosen as the loss function of the proposed FD-GRU neural network and expressed as:

$$\text{MSELoss} = \frac{1}{MN} \sum_{j=1}^M \sum_{t=1}^{N_j} \left(\text{GRU}_d \left(\left[\left(K_{ij}(t) - \text{GRU}_f(\omega_j(t), \mathbf{h}_{fj}(t-1)) \right), K_{Fj}(t) \right]^T, \mathbf{h}_{dj}(t-1) \right) \right)^2 \quad (31)$$

where M denotes the number of sequences, j denotes the No of the j -th sequence, and N_j denotes the number of sample times, i.e., the length of the j -th sequence. $y_j(t)$ denotes the label value at time t in the j -th sequence. $i_j(t)$, $\omega_j(t)$, and $K_{Fj}(t)$ denote the drive current, motor velocity and estimated motor acceleration at time t in the j -th sequence, respectively. $\mathbf{h}_{fj}(t-1)$ and $\mathbf{h}_{dj}(t-1)$ denote the hidden states of the Friction-GRU and Dynamics-GRU of the proposed FD-GRU neural network at time $t-1$ in the j -th sequence, respectively.

5. Experimental verification

5.1. Experimental setup

A set of hole milling experiments containing different combinations of cutting parameters were carried out on a DMG DMU80P machining center to verify the effectiveness and advantages of the proposed method. The experimental setup is shown in Fig. 4. The drive current and motor velocity of the X-axis and Y-axis of the machine tool are both inherent servo signals of CNC systems, and they were all obtained through TM31 modules of the used Siemens 840Dsl CNC system. TM31 modules are the additional terminal modules of the Sinamics unit in the 840Dsl system, which can provide the digital and analog in-and outputs. Since the drive current and motor velocity are the inherent analog variables in the 840Dsl system, they can be output by TM31 modules. Two motor torque constants were queried from the CNC system, where the torque constants of the X-axis motor and Y-axis motor are 2.37 Nm/A and 2.26 Nm/A, respectively. X-axis cutting force and Y-axis cutting force (i.e., cutting forces in the directions of the X-axis and Y-axis) were measured by a KISTLER 9255C dynamometer. All cutting force signals and servo signals of the X-axis and Y-axis were collected synchronously by a signal acquisition system consisting of an NI 9220 and an NI 9185. NI 9120 is an analog acquisition module that supports 16-channel synchronous acquisition. All analog signals including cutting force signals and servo signals were collected by it. NI 9185 can communicate with the host computer through TCP/IP protocol, and the collected data were obtained by a self-developed monitoring software. The sampling frequency of all signals was set to 8 kHz. The used tool is a three-fluted end mill with a diameter of 20 mm. The workpiece material is 7075 aluminum alloy.

Since the diameter of the designed hole is 90 mm and the diameter of the tool (20 mm) is much smaller than that of the hole, several circular trajectories from inside to outside are required for hole milling at each layer. The feedrates of the X-axis and Y-axis under this working condition are sine or cosine functions. On the one hand, time-varying complex X-axis and Y-axis cutting forces can be generated simultaneously. On the other hand, both X-axis and Y-axis velocities change continuously and cross the zero point, which can fully excite the nonlinear dynamic friction characteristics of the feed system. In addition, as shown in Table 1, a set of traversal combinations of cutting parameters were set up to ensure the diversity of working conditions in experiments. Among them, the spindle speed, axial depth, and radial depth of every circular toolpath are traversed and combined, and the feedrate is randomly selected from 1200 mm/min, 1400 mm/min, and 1600 mm/min. Different combinations of the spindle speed and feedrate are equivalent to different feeds per tooth. Finally, 63 trajectory data containing X-axis and Y-axis data were obtained, where 51 trajectory data were randomly

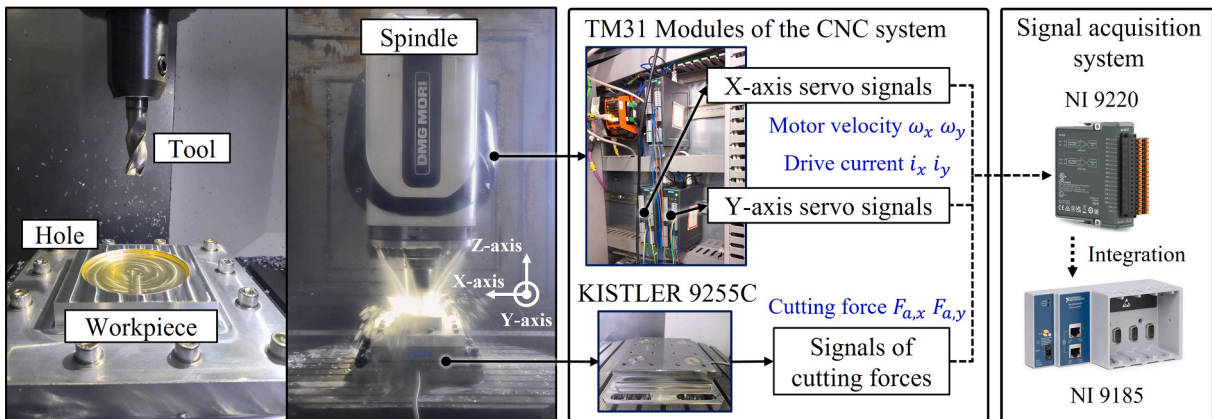


Fig. 4. Experimental setup.

Table 1
Cutting parameter settings of milling experiments.

Cutting parameter	Value
Spindle speed (r/min)	1500/1750/2000/2250/2500/2750/3000
Axial depth (mm)	2/3/4
Radial depth (mm)	5/10/15
Feedrate (mm/min)	1200/1400/1600

selected as the training dataset and the remaining 12 trajectory data were used as the test dataset.

5.2. Verification results

To verify the effectiveness and advantages of the proposed FD-GRU neural network, the GRU neural network was used for comparative verifications on the X-axis and Y-axis cutting force prediction tasks. Since the proposed FD-GRU neural network contains a Kalman filter for estimating motor acceleration, the same Kalman filter was also embedded in the GRU neural network to ensure the fairness of the comparison. Therefore, the estimated motor acceleration was selected as the motion state, and the GRU neural network used for comparison can be represented as:

$$F_a(t) = \text{GRU}_{\text{NN}}([i(t), \omega(t), \text{KF}(t)]^T, \mathbf{h}(t-1)) \quad (32)$$

It should be noted that to reduce the computational consumption of the training process, all monitored servo signals and cutting force signals were downsampled from 8 kHz to 1 kHz. Therefore, the time-discrete steps of the GRU neural network and the FD-GRU neural network were both 1 ms.

To comprehensively evaluate the model performance, Mean Absolute Error (MAE) and Relative Root Mean Square Error (RRMSE) were adopted as accuracy indices and calculated by:

$$\text{MAE} = \frac{1}{N} \sum_{j=1}^N |\hat{y}_j - y_j| \quad (33)$$

$$\text{RRMSE} = \sqrt{\frac{\sum_{i=1}^N (\hat{y}_j - y_j)^2}{\sum_{i=1}^N y_i^2}} \quad (34)$$

where N denotes the number of samples in the dataset. y_j and $\hat{y}_j$ denote the j -th label and predicted value, respectively. MAE measures the average error of the predicted value over the entire dataset, and the unit is the same as the predicted cutting force, i.e., N. RRMSE measures the relative average error of the predicted value relative to the label value over the entire dataset, and the unit is %.

To eliminate the influence of the model size on verification results and fully demonstrate the advantages of the proposed FD-GRU neural network, four sets of models with different hidden state dimensions were trained and tested for comparison as shown in Table 2 and Table 3. The hidden state dimensions of the Friction-GRU in four FD-GRU neural networks were all set to 32. The hidden state dimensions of the Dynamics-GRU in four FD-GRU neural networks were set to 192, 256, 384, and 512, respectively. To ensure the comparison fairness, the hidden state dimension of the GRU neural network was set to the sum of the hidden state dimensions of the two sub models in the corresponding FD-GRU neural network. Therefore, the hidden state dimensions of four GRU neural networks were set to 224, 288, 416, and 544, respectively. For other training settings, the epochs and batch sizes of all models were set to 50,000 and 17, respectively. The epoch was set large enough to ensure that the test loss of the trained model could reach a minimum, and the batch size was set based on the training dataset's size (51 trajectory data) and the GPU's memory (24 GB) comprehensively. The learning rates of all models were fine-tuned multiple times between 0.0001 and 0.0005, making each model achieve its best performance as much as possible. In practice, all models were trained five times using a learning rate of [0.0001, 0.0002, 0.0003, 0.0004, 0.0005], and the learning rate with the smallest test error is selected as the final setting. By the way, all

Table 2
Comparison results of the X-axis cutting force prediction task.

Dimension	Model	Training MAE (N)	RRMSE (%)	Test MAE (N)	RRMSE (%)
32 + 192	FD-GRU	16.78	9.58	20.21	11.24
224	GRU	25.31	14.19	28.29	15.10
32 + 256	FD-GRU	22.60	12.57	26.16	13.98
288	GRU	24.64	13.91	27.25	14.84
32 + 384	FD-GRU	23.64	12.97	26.83	14.36
416	GRU	31.61	18.38	32.78	17.89
32 + 512	FD-GRU	14.99	8.61	18.93	10.49
544	GRU	22.75	12.56	27.09	14.46

Table 3

Comparison results of the Y-axis cutting force prediction task.

Dimension	Model	Training MAE (N)	RRMSE (%)	Test MAE (N)	RRMSE (%)
32 + 192	FD-GRU	29.67	15.82	31.83	16.70
224	GRU	37.24	20.21	41.67	23.07
32 + 256	FD-GRU	19.22	10.72	22.41	12.64
288	GRU	32.98	17.82	35.93	20.52
32 + 384	FD-GRU	21.06	11.46	23.43	12.58
416	GRU	33.50	18.41	37.98	21.42
32 + 512	FD-GRU	22.05	12.06	25.16	14.23
544	GRU	28.66	15.56	30.80	17.19

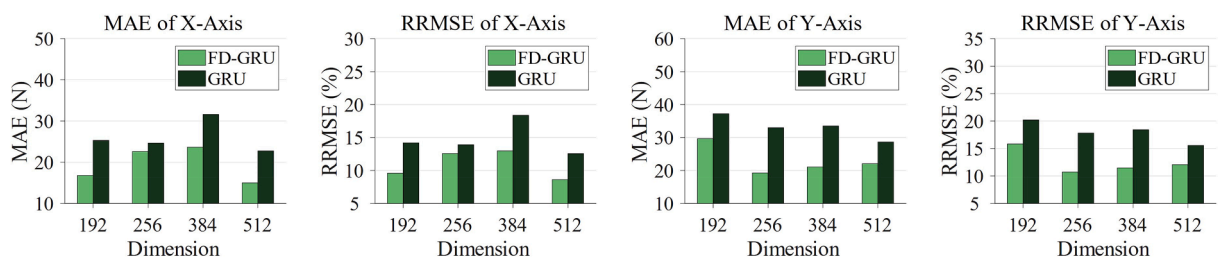
the above-mentioned procedures were carried out under a fixed random seed. The optimizers of all models were set to Adam. To avoid overfitting, the classic early stopping method was adopted, i.e., the model with the smallest test error within the entire epoch was selected as the final model.

The comparison results of X-axis and Y-axis cutting force prediction tasks are shown in Table 2 and Table 3. To show the comparison results more intuitively, the accuracy indices in Tables 2 and 3 are further plotted into histograms as shown in Fig. 5 and Fig. 6. For concise representation, the hidden state dimension of Dynamics-GRU in the FD-GRU neural network is used in Fig. 5 and Fig. 6 to distinguish different comparison groups. It is obvious that for all groups of comparisons with the same hidden state dimensions, the proposed FD-GRU neural networks have better training and test accuracy than the GRU neural network. It can be concluded that under the same training settings and model complexity, the proposed FD-GRU neural network can achieve better prediction accuracy and generalization performance than the GRU neural network.

More importantly, the worst FD-GRU neural network can even achieve higher or very similar training and test accuracy than the best GRU neural network. For the X-axis cutting force prediction task, the MAEs of the worst FD-GRU neural network (32 + 384) on the training dataset and test dataset are 23.64 N and 26.83 N respectively, and the RRMSEs are 12.97 % and 14.36 % respectively. As a comparison, the MAEs of the best GRU neural network (544) on the training dataset and test dataset are 22.75 N and 27.09 N respectively, and the RRMSEs are 12.56 % and 14.46 % respectively. For the Y-axis cutting force prediction task, the MAEs of the worst FD-GRU neural network (32 + 192) on the training dataset and test dataset are 29.67 N and 31.83 N respectively, and the RRMSEs are 15.82 % and 16.70 % respectively. As a comparison, the MAEs of the best GRU neural network (544) on the training dataset and test dataset are 28.66 N and 30.80 N respectively, and the RRMSEs are 15.56 % and 17.19 % respectively. Therefore, compared to the GRU neural network, the proposed FD-GRU neural network has better robustness to network architecture, and has higher model structure efficiency, i.e., accurate prediction can be achieved with a smaller model. These phenomena are also consistent with the theoretical analysis in Section 4.

The optimal performance of the two types of models is further compared. For the X-axis cutting force prediction task, compared with the optimal GRU neural network (544), the MAEs of the optimal FD-GRU neural network (32 + 512) on the training dataset and test dataset are decreased by 34.10 % and 30.13 % respectively, while the RRMSEs decreased by 31.43 % and 27.45 % respectively. For the Y-axis cutting force prediction task, compared with the optimal GRU neural network (544), the MAEs of the optimal FD-GRU neural network (32 + 256) on the training dataset and test dataset are decreased by 32.94 % and 27.24 % respectively, while the RRMSEs decreased by 31.10 % and 26.48 % respectively. To more comprehensively show the comparison results, the accuracy indices distribution of every trajectory data on the training and test datasets for the above-mentioned four optimal models are shown in Fig. 7. It can be concluded that with a limited number of model size optimizations, the performance upper limit of the proposed FD-GRU neural network is significantly improved compared to the GRU neural network. To further compare the training stability and eliminate the influence of random factors, four optimal models were trained again under another four random seeds without changing the other hyperparameters. The means and standard deviations of their accuracy indices are listed in Table 4. It is obvious that the proposed FD-GRU neural network has better training stability.

To compare the prediction effects more intuitively, four trajectories in the training dataset and four trajectories in the test dataset are selected for regression comparison. The cutting parameters of these eight trajectories are listed in Table 5. The optimal FD-GRU neural networks (i.e., 32 + 512 for X-axis and 32 + 256 for Y-axis) and the optimal GRU neural networks (i.e., 544 for X-axis and

**Fig. 5.** Comparison of X-axis and Y-axis cutting force prediction accuracy indices on the training dataset.

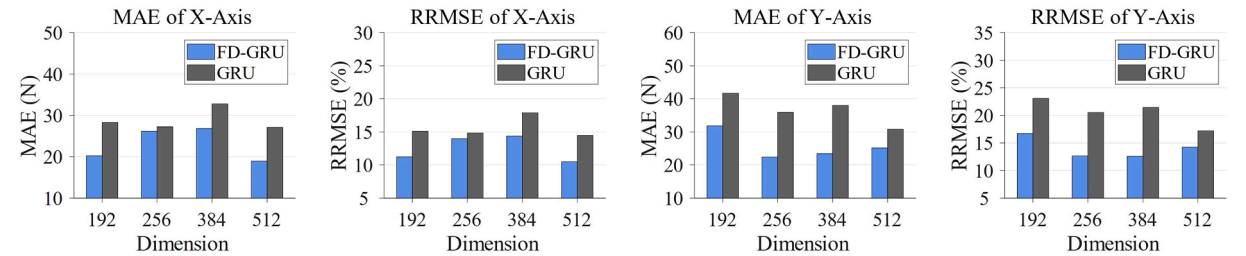


Fig. 6. Comparison of X-axis and Y-axis cutting force prediction accuracy indices on the test dataset.

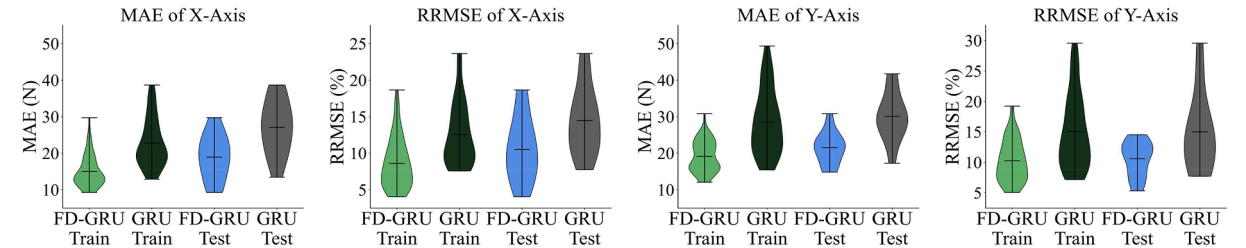


Fig. 7. Accuracy indices distribution of every trajectory data on the training and test datasets for four optimal models.

Table 4
Comparison results of the training stability of four optimal models.

Axis	Dimension	Model	Training MAE (N)	RRMSE (%)	Test MAE (N)	RRMSE (%)
X	32 + 512	FD-GRU	17.88 (2.07)	10.16 (1.10)	21.77 (2.01)	12.15 (1.20)
		GRU	28.58 (4.39)	16.27 (2.75)	31.03 (3.05)	16.81 (1.78)
Y	32 + 256	FD-GRU	20.21 (0.74)	11.15 (0.34)	23.87 (0.94)	13.35 (0.43)
		GRU	32.29 (2.56)	17.70 (1.51)	35.05 (3.01)	19.82 (1.86)

544 for Y-axis) are used for predicting the X-axis and Y-axis cutting forces in Figs. 8-11 Fig. 8 and Fig. 9 show the comparison of the prediction effects of four trajectories in the training dataset, and Fig. 10 and Fig. 11 show the comparison of the prediction effects of four trajectories in the test dataset. It can be seen that the prediction effects of FD-GRU neural networks are significantly better than GRU neural networks, whether for the training dataset or the test dataset. It should be noted that the prediction effects of GRU neural networks are not poor, but the prediction effects of FD-GRU neural networks are superior. This not only demonstrates the rationality of the definition of real-time cutting force prediction problem in Section 2, but also highlights the fairness of the comparison in this paper.

To show the excellent performance of the proposed FD-GRU neural network more intuitively, three additional trajectories (denoted as trajectory 5 ~ trajectory 7) in the test dataset are selected to directly show the predicted X-axis and Y-axis cutting forces. The cutting parameters of these three trajectories are listed in Table 5. The optimal FD-GRU neural networks (i.e., 32 + 512 for X-axis and 32 + 256 for Y-axis) are used for predicting the X-axis and Y-axis cutting forces in Figs. 12-17. For each figure, the first row of subfigures shows the cutting force on the overall trajectory, and the second and third rows of subfigures show partial zooms. All solid red lines represent

Table 5
Cutting parameters of the trajectories that are used for visualization in Figs. 8-18.

Dataset	Trajectory No	Spindle speed (r/min)	Axial depth (mm)	Radial depth (mm)	Feedrate (mm/min)
Training	1	1500	2	5	1600
	2	1750	2	5	1400
	3	2000	4	15	1200
	4	2750	4	10	1400
Test	1	1750	3	10	1400
	2	2250	4	15	1200
	3	2500	3	10	1200
	4	3000	2	15	1400
	5	1500	2	10	1400
	6	2250	3	5	1200
	7	3000	4	10	1600

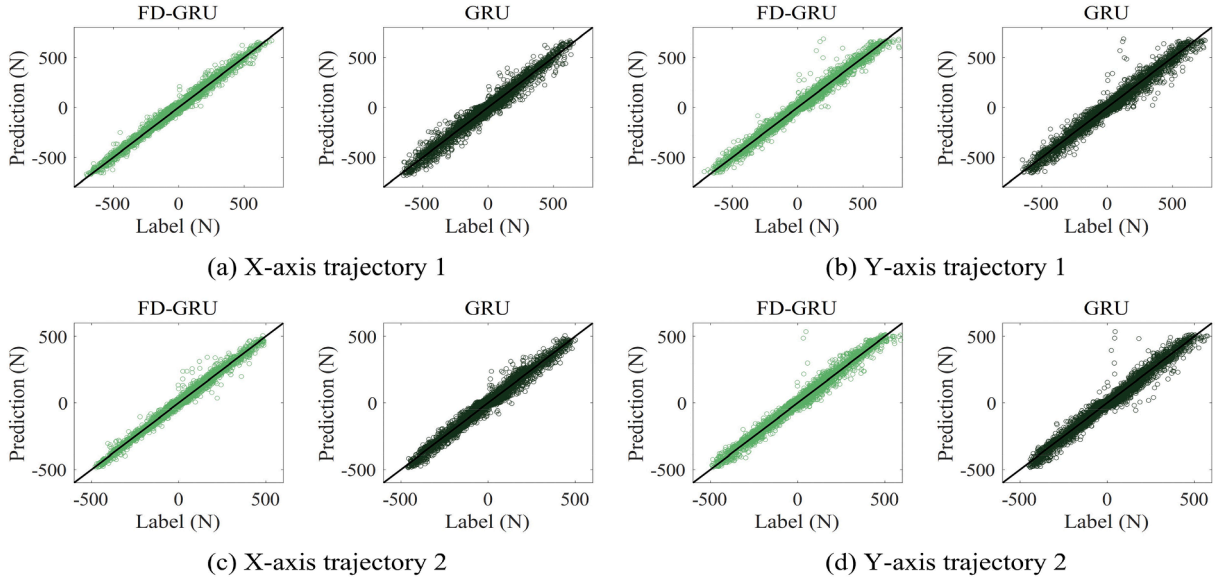


Fig. 8. Comparison of the prediction effects of trajectory 1 and trajectory 2 in the training dataset.

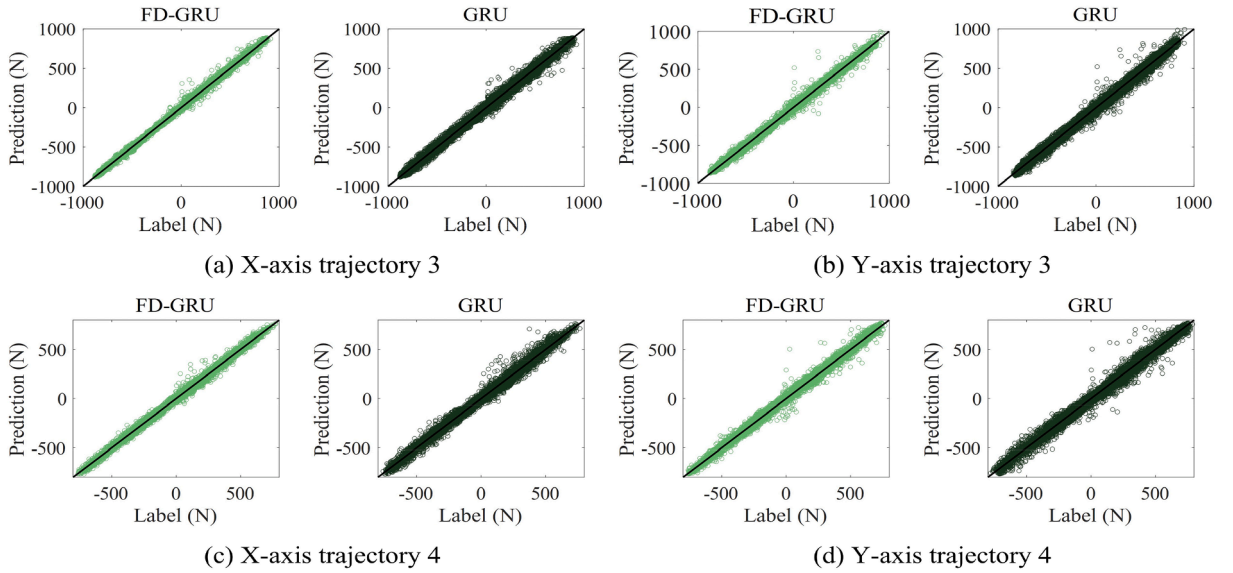


Fig. 9. Comparison of the prediction effects of trajectory 3 and trajectory 4 in the training dataset.

the predicted values, and all dashed blue lines represent the label values. Although the frequency and shape characteristics of the cutting force under the three sets of cutting parameters are quite different and the amplitude of the cutting force changes all the time, the prediction effects of cutting forces are still very satisfactory. In conclusion, the excellent prediction accuracy and generalization performance demonstrate that the proposed FD-GRU neural network has great potential to be applied in actual machining processes.

To further verify the effectiveness of the mechanism constraints in the proposed FD-GRU neural network, the monitored X-axis and Y-axis motor velocities and the estimated X-axis and Y-axis friction torques of trajectory 5 in the test dataset are shown in Fig. 18. The estimated friction torque is the output of the Friction-GRU in the FD-GRU neural network. Two used FD-GRU neural networks are still the two optimal models for X-axis ($32 + 512$) and Y-axis ($32 + 256$). It should be noted that the unit of the motor velocity is mm/min, which is the equivalent velocity of the feed axis. In addition, as stated in Section 4, although the estimated friction torque is just a relative value without units, the most important task for Friction-GRU is to effectively capture the characteristics of nonlinear dynamic friction. It is gratifying that according to the content shown in Fig. 18, it can be inferred that two Friction-GRUs have successfully learned the main friction characteristics. Specifically, when the feed axis moves at a constant velocity, the estimated friction torque also remains unchanged. In the high-velocity region far from zero, the estimated friction torque and motor velocity exhibit obvious

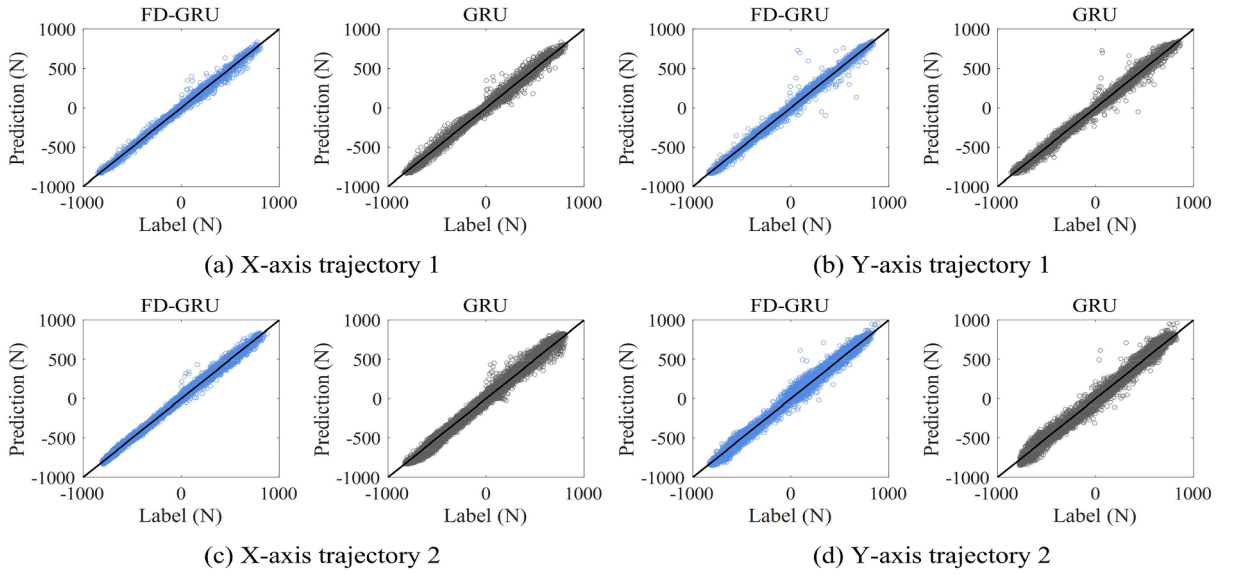


Fig. 10. Comparison of the prediction effects of trajectory 1 and trajectory 2 in the test dataset.

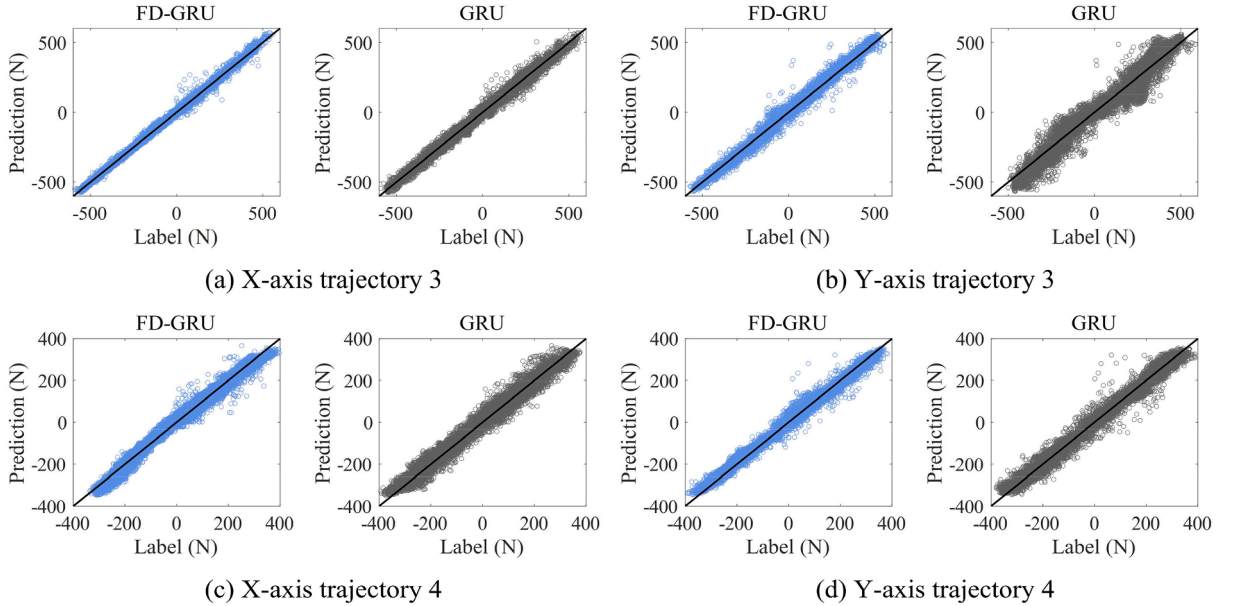


Fig. 11. Comparison of the prediction effects of trajectory 3 and trajectory 4 in the test dataset.

linear correlation characteristics, which means that viscous friction dominates. More importantly, in the low-velocity region close to zero, the estimated friction torque varies drastically but continuously, eventually changing direction with the motor velocity. These phenomena are very consistent with the characteristics of nonlinear dynamic friction, indicating that the Friction-GRU in the FD-GRU neural network can capture the nonlinear dynamic friction characteristics of the machine tool feed system very well. Since one of the two sub neural networks in the FD-GRU neural network has learned the mechanism meaning of nonlinear dynamic friction, it means that the other one, i.e., the Dynamics-GRU, also has learned its corresponding mechanism meaning. Therefore, it can be concluded that the proposed FD-GRU neural network can learn the clear mechanism meaning, which provides a strong explanation for its excellent performance.

6. Conclusion

In this paper, a novel mechanism-informed FD-GRU neural network for real-time cutting force prediction using CNC inherent servo

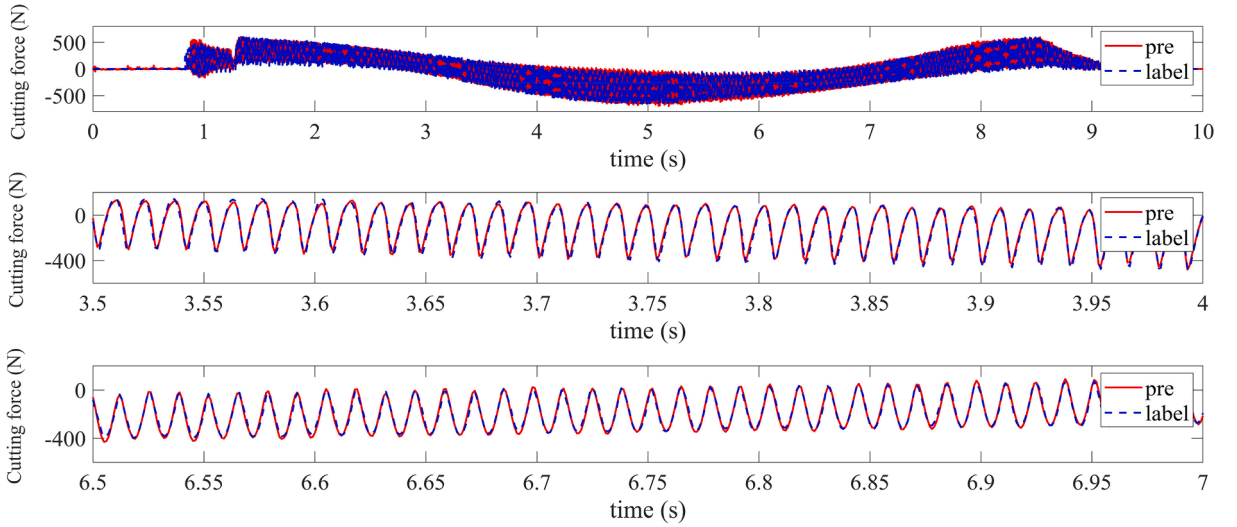


Fig. 12. Predicted X-axis cutting forces of trajectory 5 in the test dataset.

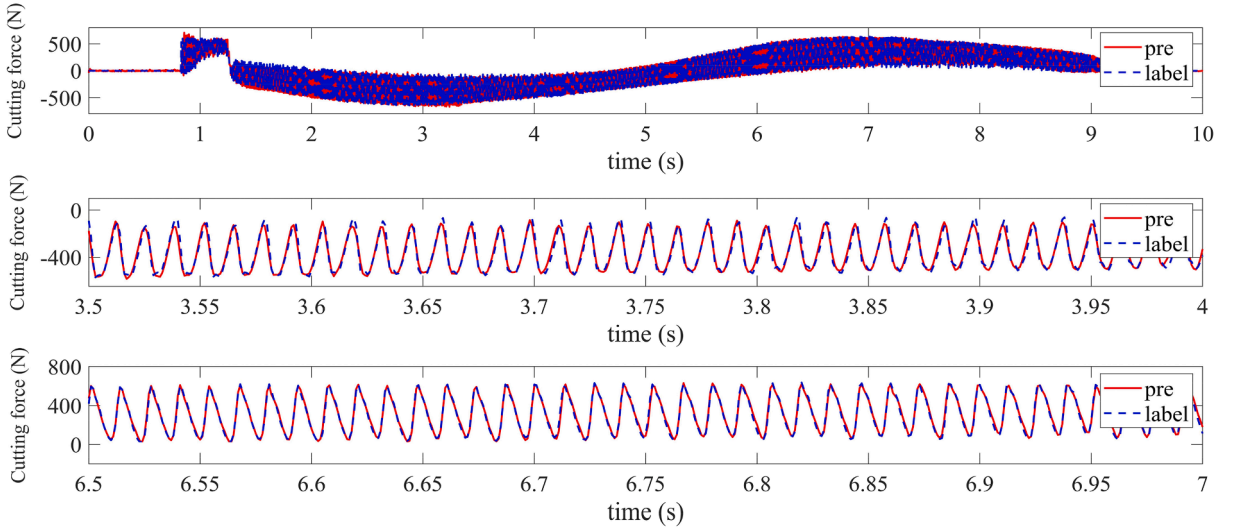


Fig. 13. Predicted Y-axis cutting forces of trajectory 5 in the test dataset.

signals is proposed. The proposed FD-GRU neural network is designed by introducing two sub GRU neural networks to represent the parts of nonlinear dynamic friction estimation and cutting force inverse estimation respectively, and structurally preserving the mechanism relationship among all related variables. In addition, a Kalman filter for estimating acceleration is embedded in the part of cutting force inverse estimation to provide more effective input information. A set of hole milling experiments with different combinations of spindle speed, axial depth, radial depth, and feedrate were carried out for the verification dataset. The main verification results can be summarized as follows:

- 1) Four sets of the GRU neural network and FD-GRU neural network with different hidden state dimensions were trained and tested for comparison. It can be concluded that under the same training settings and model complexity, the proposed FD-GRU neural network can achieve better prediction accuracy and generalization performance than the GRU neural network.
- 2) The worst FD-GRU neural network can even achieve higher or very similar training and test accuracy than the best GRU neural network. It means that the proposed FD-GRU neural network has better robustness to network architecture, and has higher model structure efficiency.
- 3) By comparing the optimal performance of two types of models, the proposed FD-GRU neural network can improve prediction accuracy and generalization performance by nearly 30 %.

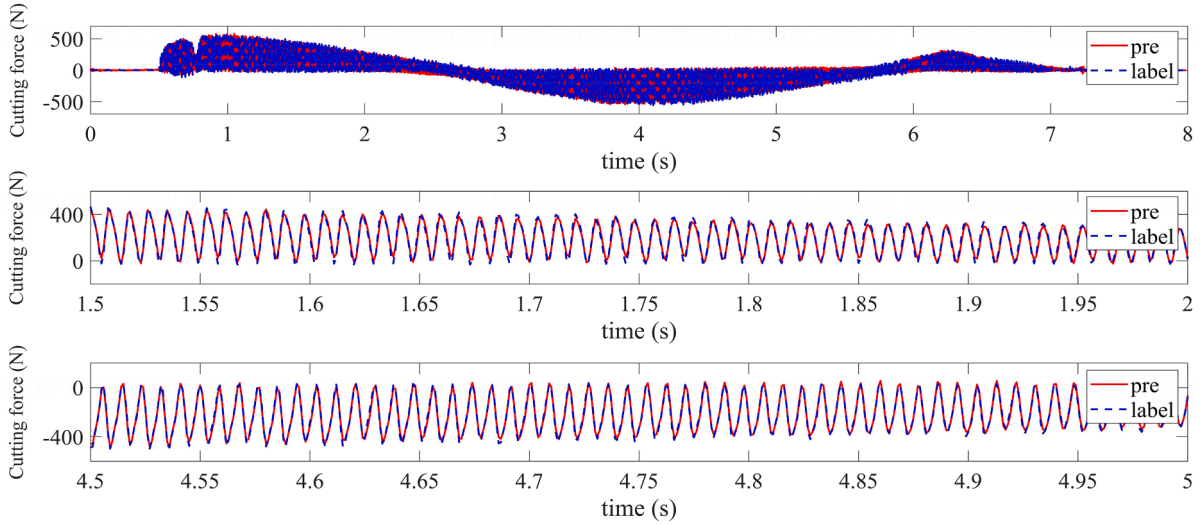


Fig. 14. Predicted X-axis cutting forces of trajectory 6 in the test dataset.

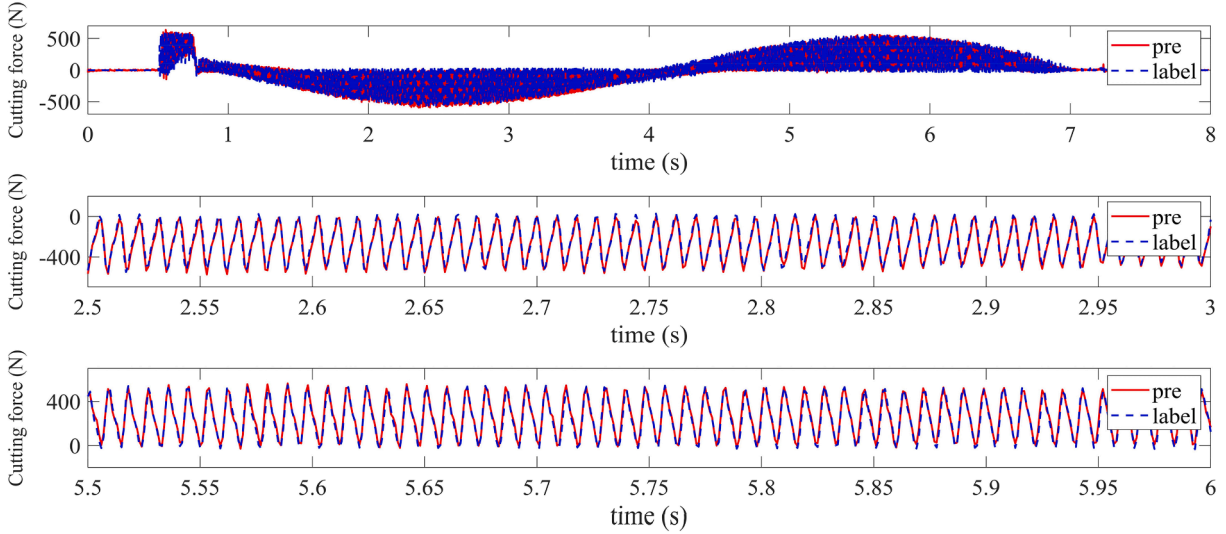


Fig. 15. Predicted Y-axis cutting forces of trajectory 6 in the test dataset.

- 4) Four optimal models were trained again under another four random seeds without changing the other hyperparameters. The experimental results have shown that the proposed FD-GRU neural network has better training stability.
- 5) The Friction-GRU in the FD-GRU neural network can capture the nonlinear dynamic friction characteristics of the machine tool feed system very well, which means the FD-GRU neural network can learn the clear mechanism meaning. It provides a strong explanation for the excellent performance of the FD-GRU neural network.

Although the proposed FD-GRU neural network has demonstrated very good performance in verification, this is only within the scope of limited tests considering the independent and identically distributed assumption of machine learning. How to use as little data outside the training distribution as possible to make the proposed FD-GRU neural network adapt well to new prediction tasks is a problem worthy of further in-depth research. In addition, due to the sequence-to-sequence training approach and the sequences are long, the computational cost of training the GRU neural network and FD-GRU neural network is relatively high. How to reduce the computational cost is also the next research direction. In future research, we will further try to combine this method with applications such as tool wear/breakage monitoring and online feedrate optimization based on cutting force, so as to achieve stable and reliable machining state monitoring and adaptive machining at a relatively lower additional cost. In addition, the proposed FD-GRU neural network has the potential to be applied for more similar friction-dynamics coupling problems, such as the prediction of servo errors of machine tools and contact forces of robots. We will further try to verify and apply it in more situations.

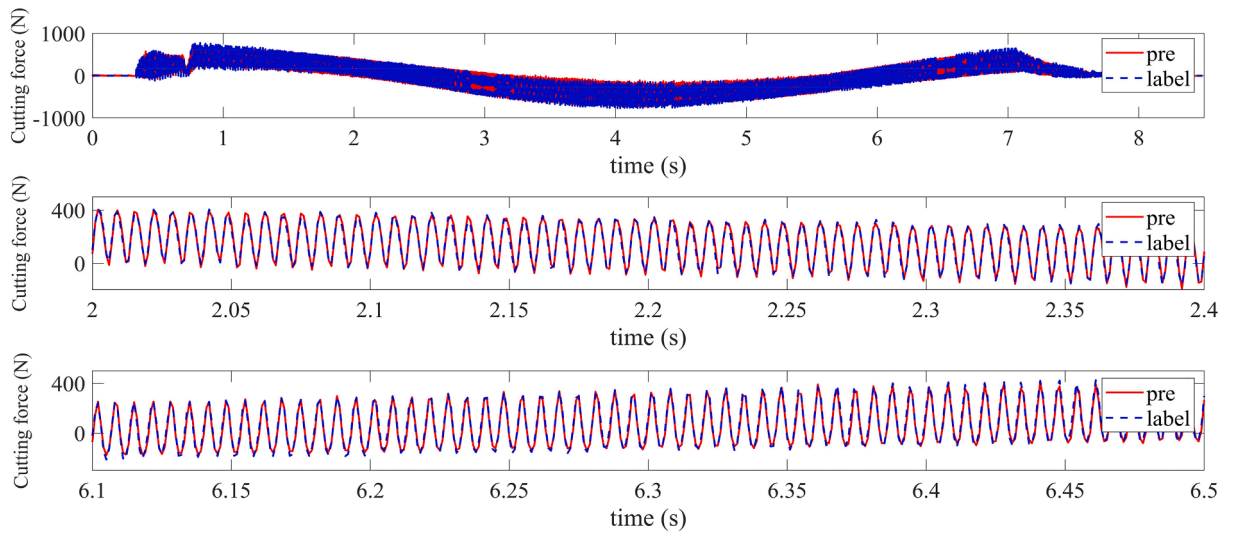


Fig. 16. Predicted X-axis cutting forces of trajectory 7 in the test dataset.

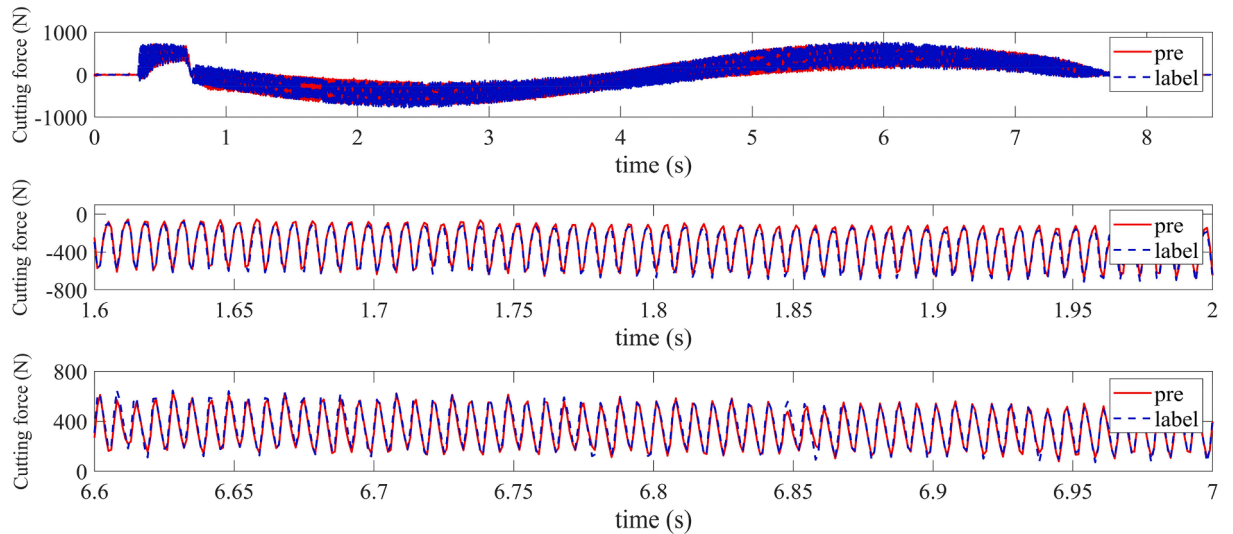


Fig. 17. Predicted Y-axis cutting forces of trajectory 7 in the test dataset.

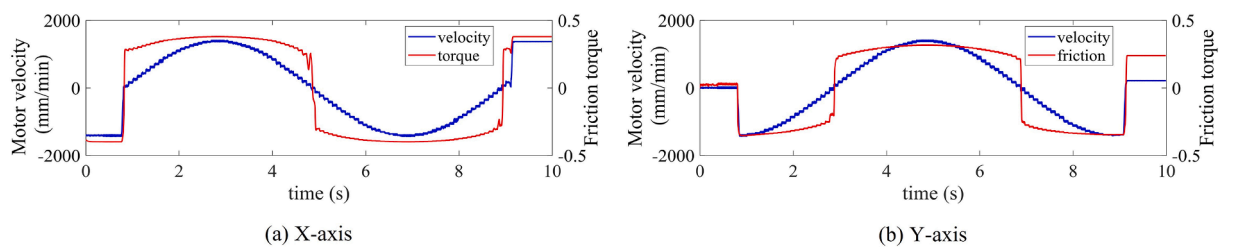


Fig. 18. Monitored motor velocities and the estimated friction torques of trajectory 5 in the test dataset.

CRediT authorship contribution statement

Yinghao Cheng: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Yingguang Li:** Writing – review & editing, Supervision, Methodology, Funding acquisition,

Conceptualization. **Qiyang Zhuang**: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation. **Xu Liu**: Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Conceptualization. **Ke Li**: Writing – original draft, Validation, Software, Methodology, Investigation. **Changqing Liu**: Writing – review & editing, Supervision, Methodology, Conceptualization. **Xiaozhong Hao**: Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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